

Junjie Yu

+86 15021875291 | errolljunjie@gmail.com | Shanghai, China

Professional Summary

- A long-term goal to produce groundbreaking results
- Continuously follow academic frontier research
- Active and creative thinker
- Value an open, free, and collaborative team culture built on candid discussion
- Skilled in project risk control and long-term planning

Education

M.S. in Mathematics — Southern University of Science and Technology (SUSTech), Shenzhen, China 2021 – 2023
Research focus: Optimization / Inverse Problem Reconstruction

B.S. in Mathematics and Applied Mathematics — Hunan Normal University, Changsha, China 2015 – 2019

Work Experience

Image Algorithm Engineer — Shenzhen Wisonic Medical Technology Co., Ltd. Jun 2023 – Apr 2025

System Algorithm Engineer — Krinwave Technology (Shenzhen) Co., Ltd. May 2025 – Jun 2026

Key Projects:

1. **Anisotropic sound-speed reconstruction** — Inverse-problem convex optimization for real-time per-point tissue sound speed; a wave-equation-constrained PINN to invert the continuous full-field speed from raw RF data; and eikonal-based time-of-flight to improve resolution and reduce artifacts.
2. **Single-focus transmit imaging** — Phase correction for focused transmit (better resolution & frame rate); derived an approximate analytic solution to the Fresnel diffraction integral; achieved whole-field imaging from a single focused transmit.
3. **Full-channel synthetic aperture inversion** — Reconstructed single-channel signals from a focused transmit; combined with sound-speed inversion to optimize focused-transmit resolution.
4. **Multiple-reflection suppression** — Suppressed vessel and near-field clutter and reduced multiple reflections.
5. **PW mode auto-optimization** — One-click auto-tuning of spectral gain, velocity scale, and baseline.
6. **Development platforms** — Non-uniform-medium simulation platform; general-purpose data-exchange platform.
7. **System-level front-end** — Built the full transmit/receive algorithm chain; improved virtual-source and multi-beam beamforming for off-focus regions.
8. **Image tuning** — Optimized transcranial B-mode and carotid-brain integrated imaging (best-in-class transcranial B-mode).

Publication

- Chao Wang, Ming Yan, Junjie Yu. "Sorted L1/L2 Minimization for Sparse Signal Recovery." Journal of Scientific Computing. (Mathematics journals list authors alphabetically.)

Graduate Research

- Applied compressed sensing to variable selection, signal denoising, and accurate sparse recovery of high-dimensional data
- Combined optimization with deep-learning priors (Plug-and-Play, PnP) for image reconstruction

Awards & Honors

- National Mathematical Contest in Modeling for College Students — Provincial Third Prize
- Interdisciplinary Contest in Modeling (ICM/MCM) — Honorable Mention
- Comprehensive Assessment Scholarship — Third Prize
- Academic Excellence Award, Hunan Normal University
- Student Science & Technology Innovation Award

Skills

- Simulation & Imaging: proficient in k-Wave and Field II; full and accurate wave-simulation and B-mode / C-mode / PW imaging capability
- Programming: MATLAB, C++, Python
- Language: CET-6 (College English Test Band 6); regularly tracks frontier literature in ultrasound imaging